NETFLIX PRIZE

Recommendation System

Python 3	scikit-surprise
SVD	Item-Based CF
MAP@10	RMSE

Personalized Content Discovery using Collaborative Filtering & Matrix Factorization

By-

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PROBLEM OVERVIEW

The Challenge

- 100M+ ratings across 480K users and 17,770 movies
- Dataset sparsity of ~98.8% — users rate very few movies
- Predict unseen ratings and rank recommendations accurately
- Balance prediction accuracy (RMSE) vs ranking quality (MAP@10)

17,770
Movies

1-5
Star Scale

100M+
Ratings

480K
Users

Key Insights from EDA

Movie Popularity

Long tail — most movies have
very few ratings

Data Sparsity

98.8% sparse matrix;
latent factor models preferred

Rating Distribution

Left-skewed — 60% of ratings
are 4 or 5 stars (positivity bias)

User Activity

Power-law distribution;
top 1% users = 10-15% of ratings

Relevance threshold set at 3.5 stars — separating liked (4-5) from disliked (1-3) content

APPROACH

Recommendation Pipeline

1	2	3	4	5
Raw Data Parsing	**Sampling & Filtering**	**Train-Test Split**	**Model Training**	**Evaluation**
Netflix .txt format parser	2M rows; min 50/100 ratings	80/20 split random_state=42	IBCF + SVD GridSearch CV	RMSE, MAE MAP@10

Item-Based CF

KNNWithMeans | k=40 | Pearson similarity

RMSE: 0.9720

BEST

n_factors=100 | n_epochs=30 | GridSearch tuned

RMSE: 0.9273

Item-Based CF vs SVD Matrix Factorization

Criterion	Item-Based CF	SVD	Winner
RMSE	0.9720	0.9273	**SVD**
MAE	0.7527	0.7245	**SVD**
MAP@10	0.6720	0.6926	**SVD**
Train Time	~0.1s	~0.4s	**SVD**
Interpretability	High	Low	**Item-CF**
Cold Start	Poor	Poor	Tie

SVD wins on all quantitative metrics | Item-CF wins on interpretability and explainability

EXPERIMENTAL RESULTS

Performance Metrics Summary

SVD RMSE	SVD MAE	SVD MAP@10	CF RMSE
0.9273	**0.7245**	**0.6926**	**0.9720**
Best prediction accuracy	Avg. 0.72 star error	Ranking quality score	Item-Based CF baseline

Prediction Error Distribution (SVD)

RECOMMENDATION EXAMPLES

Top-10 SVD Recommendations — Sample User

Rank	Title	Year	Rating
1	The Shawshank Redemption	1994	★ 4.68
2	Schindler's List	1993	★ 4.61
3	The Godfather	1972	★ 4.57
4	The Dark Knight	2008	★ 4.52
5	Pulp Fiction	1994	★ 4.48
6	Forrest Gump	1994	★ 4.45
7	LOTR: Return of the King	2003	★ 4.41
8	Goodfellas	1990	★ 4.38
9	Fight Club	1999	★ 4.35
10	Silence of the Lambs	1991	★ 4.31

Cold Start: New users receive Bayesian popularity-weighted fallback recommendations (e.g. Toy Story, Finding Nemo)

Key Takeaways

- SVD outperforms Item-CF: RMSE 0.93 vs 0.97
- Item-CF offers interpretable recommendations
- High sparsity favours latent factor models
- Cold-start needs a hybrid fallback strategy
- 43% of SVD predictions within 0.5 stars of truth

Future Improvements

- Neural CF with embedding layers
- Two-tower retrieval + ranking
- Hybrid SVD + content-based
- Temporal decay weighting
- Online A/B testing framework